

1. SYSTEM INFORMATION COMMANDS

uname:

```
akarshana2428@cloudshell:~$ uname
Linux
akarshana2428@cloudshell:~$
```

uptime:

```
akarshana2428@cloudshell:~$ uname -a
Linux cs-897017188372-default 6.6.137+ #1 SMP PREEMPT_DYNAMIC Fri May 8 09:09:05 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
akarshana2428@cloudshell:~$
```

hostname:

```
akarshana2428@cloudshell:~$ hostname
cs-897017188372-default
akarshana2428@cloudshell:~$
```

free:

```
akarshana2428@cloudshell:~$ free
              total        used         free       shared    buff/cache   available
Mem:           8138096       2466556       4377172         1148       1545028       5671540
Swap:              0              0              0
akarshana2428@cloudshell:~$
```

arch:

```
akarshana2428@cloudshell:~$ arch
x86_64
akarshana2428@cloudshell:~$
```

top:

```
top - 07:31:16 up 1:46, 0 user, load average: 0.00, 0.01, 0.00
tasks: 57 total, 1 running, 56 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.5 us, 0.3 sy, 0.0 ni, 99.0 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 7947.4 total, 4204.4 free, 2411.0 used, 1577.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 5536.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	MEM	TIME+	COMMAND
1614	akarsha+	20	0	42.6g	952212	62256	S	0.3	11.7	0:55.81	node
1	root	20	0	4324	3260	2980	S	0.0	0.0	0:00.09	bash
9	syslog	20	0	222140	3860	2500	S	0.0	0.0	0:00.05	rsyslogd
25	root	20	0	38768	28116	11008	S	0.0	0.3	0:01.68	python
26	root	20	0	6196	2716	2476	S	0.0	0.0	0:00.00	logger
68	root	10	-10	12024	3920	2804	S	0.0	0.0	0:00.06	sshd
335	root	20	0	2022092	84664	61472	S	0.0	1.0	0:01.27	dockerd
415	root	20	0	1234524	13184	8336	S	0.0	0.2	0:01.43	editor-proxy
416	root	20	0	16328	6396	5472	S	0.0	0.1	0:00.01	sudo
426	root	20	0	1226100	2684	1980	S	0.0	0.0	0:00.00	tmux-agent
456	root	20	0	1793224	45296	31896	S	0.0	0.6	0:08.49	containerd
661	root	20	0	2696	1572	1472	S	0.0	0.0	0:00.00	sleep
668	akarsha+	10	-10	8020	3644	2168	S	0.0	0.0	0:00.37	tmux
669	akarsha+	10	-10	8796	5892	4036	S	0.0	0.1	0:00.07	bash
693	root	20	0	9196	4140	3760	S	0.0	0.1	0:00.00	runuser
696	akarsha+	20	0	4324	3232	2968	S	0.0	0.0	0:00.00	sh
728	akarsha+	20	0	11.3g	165596	51048	S	0.0	2.0	0:09.73	node
1599	root	10	-10	13996	9848	8472	S	0.0	0.1	0:00.01	sshd
1601	root	10	-10	13996	9580	8204	S	0.0	0.1	0:00.02	sshd
1603	akarsha+	10	-10	14256	6480	4828	S	0.0	0.1	0:00.00	sshd
1604	akarsha+	10	-10	7340	3716	3468	S	0.0	0.0	0:00.00	bash

```

0 | 2.30 Tasks: 57, 92 sh, 6 ksh, 1 running
1 | 2.30 Load average: 0.00 0.01 0.00
Mem | 2.12G/7.7G Uptime: 01:46:30
Dsp | OK/OK

Main |
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
3857 akarshanaaz 19 -10 660 244 948 S 0.0 0.0 0:00.11 httpd -D -n cindubahal -w 621570394
1 task 20 0 324 260 2980 S 0.0 0.0 0:00.08 /bin/bash /google/scripts/cnrun.sh sleep infinity
9 syslog 20 0 216M 8860 2500 S 0.0 0.0 0:00.00 /usr/sbin/rsyslogd
10 syslog 20 0 216M 8860 2500 S 0.0 0.0 0:00.02 /usr/sbin/rsyslogd
11 syslog 20 0 216M 8860 2500 S 0.0 0.0 0:00.00 /usr/sbin/rsyslogd
12 syslog 20 0 216M 8860 2500 S 0.0 0.0 0:00.02 /usr/sbin/rsyslogd
25 root 20 0 37676 2116 11008 S 0.0 0.3 0:01.68 /usr/bin/python /usr/bin/supervisord -n --pidfile=/var/run/supervisor.pid --logfile=/var/
26 root 20 0 196 276 5476 S 0.0 0.0 0:00.00 logger -t supervisor
27 root 20 0 1024 344 320 S 0.0 0.0 0:00.06 sudo /usr/sbin/sed -p 22 -o AuthorizedKeysFile/etc/sha2/authorized_keys [listeners: 0 of 10-60 startups
335 root 20 0 1974M 8664 61472 S 0.0 1.0 0:00.21 /usr/bin/dockerd -p /var/run/docker.pid -mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
391 root 20 0 1974M 8664 61472 S 0.6 1.0 0:00.21 /usr/bin/dockerd -p /var/run/docker.pid -mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
392 root 20 0 1974M 8664 61472 S 1.0 1.0 0:00.00 /usr/bin/dockerd -p /var/run/docker.pid -mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
393 root 20 0 1974M 8664 61472 S 1.0 1.0 0:00.00 /usr/bin/dockerd -p /var/run/docker.pid -mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
394 root 20 0 1974M 8664 61472 S 1.0 1.0 0:00.00 /usr/bin/dockerd -p /var/run/docker.pid -mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
413 root 20 0 1974M 8664 61472 S 1.0 1.0 0:00.26 /usr/bin/dockerd -p /var/run/docker.pid -mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
415 root 20 0 1205M 1184 8336 S 0.2 0.2 0:00.02 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
416 root 20 0 1320 836 4472 S 0.1 0:00.01 sudo /google/devshell/tmux-agent
417 root 20 0 1205M 1184 8336 S 0.2 0.2 0:00.26 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
420 root 20 0 1205M 1184 8336 S 0.2 0.2 0:00.35 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
421 root 20 0 1205M 1184 8336 S 0.2 0.2 0:00.29 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
422 root 20 0 1205M 1184 8336 S 0.2 0.2 0:00.29 /google/devshell/editor/proxy/editor-proxy -static-content-dir=/google/devshell/editor/code-oss-for-cloud-shell/static-content
426 root 20 0 1197M 684 980 S 0.0 0.0 0:00.00 /google/devshell/tmux-agent
433 root 20 0 1197M 684 980 S 0.0 0.0 0:00.00 /google/devshell/tmux-agent
434 root 20 0 1197M 684 980 S 0.0 0.0 0:00.00 /google/devshell/tmux-agent
435 root 20 0 1197M 684 980 S 0.0 0.0 0:00.00 /google/devshell/tmux-agent
456 root 20 0 1751M 4296 31896 S 0.0 0.6 0:00.05 containedd --config /var/run/docker/containers/containerd.toml
457 root 20 0 1974M 8664 61472 S 1.0 1.0 0:00.00 /usr/bin/dockerd -p /var/run/docker.pid -mtu=1460 --registry-mirror=https://asia-mirror.gcr.io
459 root 20 0 1751M 4296 31896 S 0.6 0.6 0:00.05 containedd --config /var/run/docker/containers/containerd.toml

```

```
akarshana2428@cloudshell:~$ ls
README-cloudshell.txt
akarshana2428@cloudshell:~$
```

```
akarshana2428@cloudshell:~$ mkdir mlops
akarshana2428@cloudshell:~$
```

```
akarshana2428@cloudshell:~$ pwd mlops
/home/akarshana2428
akarshana2428@cloudshell:~$
```

```
akarshana2428@cloudshell:~$ cd mlops
akarshana2428@cloudshell:~/mlops$
```

```
akarshana2428@cloudshell:~/mlops$ cd ..
akarshana2428@cloudshell:~$ rmdir mlops
akarshana2428@cloudshell:~$
```

6.TOUCH:

```
akarshana2428@cloudshell:~$ touch mlops
```

7.CP:

```
akarshana2428@cloudshell:~$ cp mlops mlops_copy
akarshana2428@cloudshell:~$ ls
mlops  mlops_copy  README-cloudshell.txt
```

8.RM:

```
akarshana2428@cloudshell:~$ rm mlops_new
akarshana2428@cloudshell:~$ ls
link1  link2  mlops  README-cloudshell.txt
```

9.MV:

```
akarshana2428@cloudshell:~$ mv mlops_copy mlops_new
akarshana2428@cloudshell:~$ ls
mlops  mlops new  README-cloudshell.txt
```

10.LN:

```
akarshana2428@cloudshell:~$ ln mlops link1
akarshana2428@cloudshell:~$ ln -s mlops link2
akarshana2428@cloudshell:~$ ls -l
total 4
-rw-rw-r-- 2 akarshana2428 akarshana2428  0 May 14 09:43 link1
lrwxrwxrwx 1 akarshana2428 akarshana2428  5 May 14 09:50 link2 -> mlops
-rw-rw-r-- 2 akarshana2428 akarshana2428  0 May 14 09:43 mlops
-rw-rw-r-- 1 akarshana2428 akarshana2428  0 May 14 09:44 mlops_new
-rwxr-xr-x 1 akarshana2428 akarshana2428 913 May 14 08:56 README-cloudshell.txt
```

3.FILE VIEWING AND EDITING

```
akarshana2428@cloudshell:~$ echo "Hello Linux
This is line 2
This is line 3
End of file" > mlops.txt
```

CAT:

```
akarshana2428@cloudshell:~$ cat mlops.txt
Hello Linux
This is line 2
This is line 3
End of file
```

MORE:

```
akarshana2428@cloudshell:~$ more mlops.txt
Hello Linux
This is line 2
This is line 3
End of file
```

LESS:

```
Hello Linux
This is line 2
This is line 3
End of file
mlops.txt (END)
```

TAIL:

```
akarshana2428@cloudshell:~$ tail mlops.txt
Hello Linux
This is line 2
This is line 3
End of file
akarshana2428@cloudshell:~$ tail -n 2 mlops.txt
This is line 3
End of file
```

HEAD:

```
akarshana2428@cloudshell:~$ head mlops.txt
Hello Linux
This is line 2
This is line 3
End of file
akarshana2428@cloudshell:~$ head -n 2 mlops.txt
Hello Linux
This is line 2
```

NANO:

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini
GNU nano 7.2 New Buffer

GNU nano 7.2 New Buffer

GNU nano 7.2 mlops.txt
Hello Linux
This is line 2
This is line 3
End of file
```

VI/VIM:

```
akarshana2428@cloudshell:~$ echo "Hello Linux This is the vim command ">ml.txt
akarshana2428@cloudshell:~$ vim ml.txt
akarshana2428@cloudshell:~$ cat ml.txt
This is the linux world
```

AWK:

```
akarshana2428@cloudshell:~$ awk '{print $1}' mlops.txt
Hello
End
akarshana2428@cloudshell:~$
```

SED:

```
akarshana2428@cloudshell:~$ sed 's/Linux/LINUX/g' mlops.txt
Hello LINUX This is the vim command3
End of file
akarshana2428@cloudshell:~$
```

4.FILE PERMISSIONS AND OWNERSHIP

1.CHMOD:

```
akarshana2428@cloudshell:~$ chmod 777 ml.txt
akarshana2428@cloudshell:~$ ls -l ml.txt
-rwxrwxrwx 1 akarshana2428 akarshana2428 24 May 14 16:12 ml.txt
```

2. CHOWN

```
akarshana2428@cloudshell:~$ sudo chown $USER:$USER ml.txt
akarshana2428@cloudshell:~$ ls -l ml.txt
-rwxrwxrwx 1 akarshana2428 akarshana2428 24 May 14 16:12 ml.txt
```

3.CHGRP:

```
akarshana2428@cloudshell:~$ sudo chgrp $USER ml.txt
akarshana2428@cloudshell:~$ ls -l ml.txt
-rwxrwxrwx 1 akarshana2428 akarshana2428 24 May 14 16:12 ml.txt
```

4.UMASK:

```
akarshana2428@cloudshell:~$ umask
0002
akarshana2428@cloudshell:~$ umask 022
```

5. STAT:

```
akarshana2428@cloudshell:~$ stat ml.txt
  File: ml.txt
  Size: 24          Blocks: 8          IO Block: 4096   regular file
Device: 8,17      Inode: 131377       Links: 1
Access: (0777/-rwxrwxrwx)  Uid: ( 1000/akarshana2428)   Gid: ( 1000/akarshana2428)
Access: 2026-05-14 16:12:21.992814224 +0000
Modify: 2026-05-14 16:12:21.992814224 +0000
Change: 2026-05-14 16:14:58.863954687 +0000
 Birth: 2026-05-14 10:40:55.343848303 +0000
```

6.GETFACL:

```
akarshana2428@cloudshell:~$ getfacl ml.txt
# file: ml.txt
# owner: akarshana2428
# group: akarshana2428
user::rwx
group::rwx
other::rwx
```

7.SETFACL:

```
akarshana2428@cloudshell:~$ setfacl -m u:$USER:r ml.txt
akarshana2428@cloudshell:~$ getfacl ml.txt
# file: ml.txt
# owner: akarshana2428
# group: akarshana2428
user::rwx
user:akarshana2428:r--
group::rwx
mask::rwx
other::rwx
```

8.LSATTR:

```
akarshana2428@cloudshell:~$ lsattr ml.txt
----i-----e----- ml.txt
```

9.CHATTR:

```
akarshana2428@cloudshell:~$ sudo chattr +i ml.txt
akarshana2428@cloudshell:~$ lsattr ml.txt
----i-----e----- ml.txt
akarshana2428@cloudshell:~$ rm ml.txt
rm: cannot remove 'ml.txt': Operation not permitted
akarshana2428@cloudshell:~$ sudo chattr -i ml.txt
akarshana2428@cloudshell:~$ rm ml.txt
```

10.USERMOD:

```
akarshana2428@cloudshell:~$ sudo usermod -aG sudo $USER
akarshana2428@cloudshell:~$ groups
akarshana2428 adm sudo docker
```

5. NETWORKING COMMANDS

1.IFCONFIG:

```
akarshana2428@cloudshell:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 36:9c:46:e6:5c:bf txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.4 netmask 255.255.0.0 broadcast 10.88.255.255
    ether ca:95:8b:76:05:f1 txqueuelen 1000 (Ethernet)
    RX packets 21065 bytes 7836855 (7.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 17692 bytes 20485727 (20.4 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 14653 bytes 40437054 (40.4 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 14653 bytes 40437054 (40.4 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

2. IP:

```
akarshana2428@cloudshell:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0@if8: <BROADCAST,MULTICAST,UP,LOWER UP> mtu 1460 qdisc noqueue state UP group default qlen 1000
    link/ether ca:95:8b:76:05:f1 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 10.88.0.4/16 brd 10.88.255.255 scope global eth0
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1460 qdisc noqueue state DOWN group default
    link/ether 36:9c:46:e6:5c:bf brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
```

3. PING

```
akarshana2428@cloudshell:~$ ping google.com
PING google.com (64.233.170.139) 56(84) bytes of data:
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=1 ttl=114 time=0.460 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=2 ttl=114 time=0.279 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=3 ttl=114 time=0.278 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=4 ttl=114 time=0.315 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=5 ttl=114 time=0.385 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=6 ttl=114 time=0.333 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=7 ttl=114 time=0.350 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=8 ttl=114 time=0.371 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=9 ttl=114 time=0.355 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=10 ttl=114 time=0.312 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=11 ttl=114 time=0.352 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=12 ttl=114 time=0.267 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=13 ttl=114 time=0.307 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=14 ttl=114 time=0.316 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=15 ttl=114 time=0.317 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=16 ttl=114 time=0.290 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=17 ttl=114 time=0.312 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=18 ttl=114 time=0.324 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=19 ttl=114 time=0.339 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=20 ttl=114 time=0.352 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=21 ttl=114 time=0.318 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=22 ttl=114 time=0.330 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=23 ttl=114 time=0.359 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=24 ttl=114 time=0.298 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=25 ttl=114 time=0.315 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=26 ttl=114 time=0.316 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=27 ttl=114 time=0.295 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=28 ttl=114 time=0.272 ms
64 bytes from sg-in-f139.1e100.net (64.233.170.139): icmp_seq=29 ttl=114 time=0.229 ms
^C
--- google.com ping statistics ---
29 packets transmitted, 29 received, 0% packet loss, time 28653ms
rtt min/avg/max/mdev = 0.229/0.322/0.460/0.042 ms
```

4. NETSTAT:

```
akarshana2428@cloudshell:~$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.1:37899         0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:971             0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:8998          0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:35193         0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:43353        0.0.0.0:*               LISTEN
tcp6       0      0 :::22                   :::*                    LISTEN
tcp6       0      0 :::970                   :::*                    LISTEN
tcp6       0      0 :::980                   :::*                    LISTEN
tcp6       0      0 :::981                   :::*                    LISTEN
tcp6       0      0 :::900                   :::*                    LISTEN
tcp6       0      0 :::922                   :::*                    LISTEN
```

5. SS:

```
akarshana2428@cloudshell:~$ ss -tuln
Netid      State      Recv-Q     Send-Q       Local Address:Port      Peer Address:Port      Process
tcp        LISTEN     0           128          127.0.0.1:37899         0.0.0.0:*
tcp        LISTEN     0           128          0.0.0.0:22              0.0.0.0:*
tcp        LISTEN     0           511         0.0.0.0:971             0.0.0.0:*
tcp        LISTEN     0           5          127.0.0.1:8998          0.0.0.0:*
tcp        LISTEN     0           511         127.0.0.1:35193         0.0.0.0:*
tcp        LISTEN     0           511         127.0.0.1:43353        0.0.0.0:*
tcp        LISTEN     0           128          [::]:22                 [::]:*
tcp        LISTEN     0           4096         *:970                   *:~
tcp        LISTEN     0           4096         *:980                   *:~
tcp        LISTEN     0           4096         *:981                   *:~
tcp        LISTEN     0           4096         *:900                   *:~
tcp        LISTEN     0           4096         *:922                   *:~
```

6. TRACEROUTE:

```
akarshana2428@cloudshell:~$ traceroute google.com
traceroute to google.com (64.233.170.138), 30 hops max, 60 byte packets
 1  10.88.0.1 (10.88.0.1)  0.054 ms  0.016 ms  0.012 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  sg-in-f138.1e100.net (64.233.170.138)  0.271 ms  * *
```

7.WGET:

```
akarshana2428@cloudshell:~$ wget https://example.com
--2026-05-14 16:49:58-- https://example.com/
Resolving example.com (example.com)... 104.20.23.154, 172.66.147.243, 2606:4700:10::ac42:93f3, ...
Connecting to example.com (example.com)[104.20.23.154]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'

index.html                               [ <=> ]          528  --.-KB/s   in 0s

2026-05-14 16:49:59 (89.4 MB/s) - 'index.html' saved [528]
```

8.CURL:

```
akarshana2428@cloudshell:~$ curl https://example.com
<!doctype html><html lang="en"><head><title>Example Domain</title><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>
```

9.NSLOOKUP:

```
akarshana2428@cloudshell:~$ nslookup google.com
Server:      169.254.169.254
Address:     169.254.169.254#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.194.101
Name:   google.com
Address: 172.217.194.113
Name:   google.com
Address: 172.217.194.100
Name:   google.com
Address: 172.217.194.138
Name:   google.com
Address: 172.217.194.102
Name:   google.com
Address: 172.217.194.139
Name:   google.com
Address: 2404:6800:4003:c01::8a
Name:   google.com
Address: 2404:6800:4003:c01::65
Name:   google.com
Address: 2404:6800:4003:c01::64
Name:   google.com
Address: 2404:6800:4003:c01::8b
```

10.DIG:

```
akarshana2428@cloudshell:~$ dig google.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; -->HEADER<<- opcode: QUERY, status: NOERROR, id: 8986
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                272     IN      A      172.217.194.101
google.com.                272     IN      A      172.217.194.113
google.com.                272     IN      A      172.217.194.100
google.com.                272     IN      A      172.217.194.138
google.com.                272     IN      A      172.217.194.102
google.com.                272     IN      A      172.217.194.139

;; Query time: 1 msec
;; SERVER: 169.254.169.254#53(169.254.169.254) (UDP)
;; WHEN: Thu May 14 16:51:31 UTC 2026
;; MSG SIZE rcvd: 135
```

6. PROCESS MANAGEMENT

PS:

```
akarshana2428@cloudshell:~$ ps
  PID TTY          TIME CMD
 2135 pts/6    00:00:00 bash
 2503 pts/6    00:00:00 ps
```


PGREP:

```
akarshana2428@cloudshell:~$ pgrep sleep
650
```

PKILL:

```
akarshana2428@cloudshell:~$ pkill firefox
akarshana2428@cloudshell:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.0   4324   3296 ?        Ss   16:04   0:00 /bin/bash /google/scripts/onrun.sh sleep infinity
syslog      9  0.0  0.0  222140  3760 ?        Ss1  16:04   0:00 /usr/sbin/rsyslogd
root       25  0.0  0.3  38768  27928 ?        S    16:04   0:01 /usr/bin/python /usr/bin/supervisord -n -c /google/devshell/
root       26  0.0  0.0   6196   2668 ?        S    16:04   0:00 logger -t supervisord
root       67  0.0  0.0  12024   4088 ?        S<s  16:04   0:00 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/etc/ssh/ke
root      320  0.0  1.0 2021836  84436 ?        Sl   16:04   0:00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 --registr
root      422  0.0  0.1 1234268 12356 ?        Sl   16:04   0:00 /google/devshell/editor/proxy/editor-proxy -static-content-d
```

KILL:

```
akarshana2428@cloudshell:~$ kill 2547
[1]+  Terminated                  sleep 300
```

KILLALL:

```
akarshana2428@cloudshell:~$ killall sleep
sleep(650): Operation not permitted
[1]+  Terminated                  sleep 300
```

FG:

```
akarshana2428@cloudshell:~$ fg
sleep 300
```

JOBS:

```
akarshana2428@cloudshell:~$ jobs
[1]+  Stopped                  sleep 300
```

NICE:

```
akarshana2428@cloudshell:~$ nice -n 10 sleep 100
```

RENICE:

```
akarshana2428@cloudshell:~$ renice 10 -p 2591
2591 (process ID) old priority -10, new priority 10
```

7. DISK MANAGEMENT

DF:

```
akarshana2428@cloudshell:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay          95G   73G   22G   78% /
tmpfs            64M    0    64M    0% /dev
/dev/sda1        95G   73G   22G   78% /root
/dev/disk/by-id/google-home-part1 4.8G  150M   4.4G    4% /home
/dev/root        2.0G   1.3G   715M   64% /usr/lib/modules
shm              64M    0    64M    0% /dev/shm
```

DU:

```
akarshana2428@cloudshell:~$ du -sh .  
150M  .
```

FDISK:

```
Disk /dev/dm-0: 1.94 GiB, 2087714816 bytes, 509696 sectors  
Units: sectors of 1 * 4096 = 4096 bytes  
Sector size (logical/physical): 4096 bytes / 4096 bytes  
I/O size (minimum/optimal): 4096 bytes / 4096 bytes  
  
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors  
Disk model: PersistentDisk  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 4096 bytes  
I/O size (minimum/optimal): 4096 bytes / 4096 bytes  
Disklabel type: dos  
Disk identifier: 0x5245a676  
  
Device      Boot Start      End  Sectors  Size Id Type  
/dev/sdb1                2048 10485759 10483712   5G 83 Linux
```

MKFS:

```
akarshana2428@cloudshell:~$ mkfs.ext4 test.img  
mke2fs 1.47.0 (5-Feb-2023)  
Discarding device blocks: done  
Creating filesystem with 12800 4k blocks and 12800 inodes  
  
Allocating group tables: done  
Writing inode tables: done  
Creating journal (1024 blocks): done  
Writing superblocks and filesystem accounting information: done
```

UMOUNT:

```
akarshana2428@cloudshell:~$ sudo umount mydisk
```

BLKID:

```
akarshana2428@cloudshell:~$ sudo blkid test.img  
test.img: UUID="c3bab8c8-e592-4ca2-82fd-a2adcbf55a24" BLOCK_SIZE="4096" TYPE="ext4"
```

LSBLK:

```
akarshana2428@cloudshell:~$ lsblk  
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
sda          8:0    0 100G  0 disk  
├─sda1       8:1    0 95.8G  0 part /run/google/devshell  
│           │           │           │ /var/lib/docker  
│           │           │           │ /etc/ssh/keys  
│           │           │           │ /var/config/tmux  
│           │           │           │ /etc/resolv.conf  
│           │           │           │ /etc/hostname  
│           │           │           │ /etc/hosts  
│           │           │           │ /root  
├─sda2       8:2    0  16M  0 part  
├─sda3       8:3    0   2G  0 part  
├─sda4       8:4    0  16M  0 part  
├─sda5       8:5    0   2G  0 part  
├─sda6       8:6    0 512B  0 part  
├─sda7       8:7    0 512B  0 part  
├─sda8       8:8    0  16M  0 part  
├─sda9       8:9    0 512B  0 part  
├─sda10      8:10   0 512B  0 part  
├─sda11      8:11   0   8M  0 part  
├─sda12      8:12   0  32M  0 part  
sdb          8:16    0   5G  0 disk  
└─sdb1       8:17    0   5G  0 part /home
```

TUNE2FS:

```
akarshana2428@cloudshell:~$ sudo tune2fs -l test.img
tune2fs 1.47.0 (5-Feb-2023)
Filesystem volume name: <none>
Last mounted on: <not available>
Filesystem UUID: c3bab8c8-e592-4ca2-82fd-a2adcbf55a24
Filesystem magic number: 0xEF53
Filesystem revision #: 1 (dynamic)
Filesystem features: has_journal ext_attr resize_inode dir_index filetype extent 64bit flex_bg sparse_super large_file huge_file dir_nlink
extra_isize metadata_csum
Filesystem flags: signed_directory_hash
Default mount options: user_xattr acl
Filesystem state: clean
Errors behavior: Continue
Filesystem OS type: Linux
Inode count: 12800
```

8. SYSTEM MONITORING AND PERFORMANCE

TOP:

```
top - 17:37:40 up 3:09, 0 user, load average: 0.07, 0.02, 0.06
Tasks: 56 total, 1 running, 55 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.8 us, 0.5 sy, 0.0 ni, 98.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7947.4 total, 4273.4 free, 2338.1 used, 1581.2 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 5609.3 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR   S  %CPU  %MEM    TIME+  COMMAND
  911 akarsha+  20   0   11.3g   106176 50948 S   0.3   1.3   0:05.71 node
 1451 akarsha+  20   0    42.6g 954508 61852 S   0.3  11.7   0:52.59 node
    1 root       20   0    4324    3296   3020 S   0.0   0.0   0:00.06 bash
    9 syslog    20   0   222140  3760   2392 S   0.0   0.0   0:00.02 rsyslogd
   25 root     20   0   38768   27928  10820 S   0.0   0.3   0:01.44 python
   26 root     20   0    6196    2668   2432 S   0.0   0.0   0:00.00 logger
   67 root     10 -10   12024    4088   2972 S   0.0   0.1   0:00.01 sshd
  320 root     20   0  2021836 84472  61216 S   0.0   1.0   0:01.08 dockerd
  422 root     20   0  1234268 12324   8264 S   0.0   0.2   0:01.23 editor-proxy
  424 root     20   0   16328    6440   5516 S   0.0   0.1   0:00.01 sudo
  445 root     20   0  1226100  2716   1988 S   0.0   0.0   0:00.01 tmux-agent
  457 root     20   0  1793224 45892  32452 S   0.0   0.6   0:06.67 containerd
  650 root     20   0    2696    1648   1552 S   0.0   0.0   0:00.00 sleep
  652 root     10 -10   13996    9600   8220 S   0.0   0.1   0:00.02 sshd
  660 root     10 -10   13996    9688   8316 S   0.0   0.1   0:00.02 sshd
  665 akarsha+  10 -10   14256    6252   4592 S   0.0   0.1   0:00.00 sshd
  666 akarsha+  10 -10    7340    3816   3568 S   0.0   0.0   0:00.00 bash
  669 akarsha+  10 -10    7604    4500   3892 S   0.0   0.1   0:00.00 bash
  672 akarsha+  10 -10   14256    6292   4640 S   0.0   0.1   0:00.03 sshd
  683 akarsha+  10 -10    7340    3780   3532 S   0.0   0.0   0:00.00 bash
  684 akarsha+  10 -10    7340    3808   3544 S   0.0   0.0   0:00.00 start-shell.sh
```

HTOP:

```
0[ ] 1.3% Tasks: 56, 92 thr, 0 kthr: 1 running
1[ ] 1.3% Load average: 0.50 0.14 0.09
Mem[ ] 2.05G/7.76G Uptime: 03:10:11
Swp[ ] OK/0K

Main CPU
PID USER      PRI  NI    VIRT    RES    SHR   S  %CPU  %MEM    TIME+  Command
3594 akarshana2 10 -10 3536 4865 3672 S   0.3   0.1   3:20.39 htop
    1 root       20   0    4324    3296   3020 S   0.0   0.0   0:00.06 /bin/bash /google/scripts/onrun.sh sleep infinity
   11 syslog    20   0    216M   3760   2392 S   0.0   0.0   0:00.00 /usr/sbin/rsyslogd
   12 syslog    20   0    216M   3760   2392 S   0.0   0.0   0:00.00 /usr/sbin/rsyslogd
   25 root     20   0   38768   27928  10820 S   0.0   0.3   0:01.45 /usr/bin/python /usr/bin/supervisord -n -c /google/d
   26 root     20   0    6196    2668   2432 S   0.0   0.0   0:00.00 logger -t supervisord
   67 root     10 -10   12024    4088   2972 S   0.0   0.1   0:00.01 sshd: /usr/sbin/sshd -p 22 -o AuthorizedKeysFile=/et
  320 root     20   0  1974M 84472  61216 S   0.0   1.0   0:00.12 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 -
  392 root     20   0  1974M 84472  61216 S   0.0   1.0   0:00.15 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 -
  393 root     20   0  1974M 84472  61216 S   0.0   1.0   0:00.01 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 -
  394 root     20   0  1974M 84472  61216 S   0.0   1.0   0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 -
  395 root     20   0  1974M 84472  61216 S   0.0   1.0   0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 -
  422 root     20   0  1205M 12324  8264 S   0.0   0.2   0:00.01 /google/devshell/editor/proxy/editor-proxy -static-c
  424 root     20   0  16328    6440   5516 S   0.0   0.1   0:00.01 sudo /google/devshell/tmux-agent
  432 root     20   0  1205M 12324  8264 S   0.0   0.2   0:00.24 /google/devshell/editor/proxy/editor-proxy -static-c
  434 root     20   0  1205M 12324  8264 S   0.0   0.2   0:00.23 /google/devshell/editor/proxy/editor-proxy -static-c
  435 root     20   0  1205M 12324  8264 S   0.0   0.2   0:00.36 /google/devshell/editor/proxy/editor-proxy -static-c
  445 root     20   0  1197M  2716   1988 S   0.0   0.0   0:00.00 /google/devshell/tmux-agent
  451 root     20   0  1197M  2716   1988 S   0.0   0.0   0:00.00 /google/devshell/tmux-agent
  452 root     20   0  1197M  2716   1988 S   0.0   0.0   0:00.00 /google/devshell/tmux-agent
  453 root     20   0  1197M  2716   1988 S   0.0   0.0   0:00.00 /google/devshell/tmux-agent
  454 root     20   0  1205M 12324  8264 S   0.0   0.2   0:00.36 /google/devshell/editor/proxy/editor-proxy -static-c
  455 root     20   0  1974M 84472  61216 S   0.0   1.0   0:00.09 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 -
  457 root     20   0  1751M 45892  32452 S   0.0   0.6   0:00.05 containerd --config /var/run/docker/containerd/conta
  458 root     20   0  1974M 84472  61216 S   0.0   1.0   0:00.15 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 -
  459 root     20   0  1751M 45892  32452 S   0.0   0.6   0:01.89 containerd --config /var/run/docker/containerd/conta
  459 root     20   0  1751M 45892  32452 S   0.0   0.6   0:01.90 containerd --config /var/run/docker/containerd/conta
  460 root     20   0  1751M 45892  32452 S   0.0   0.6   0:00.00 containerd --config /var/run/docker/containerd/conta
  461 root     20   0  1751M 45892  32452 S   0.0   0.6   0:01.58 containerd --config /var/run/docker/containerd/conta
  462 root     20   0  1751M 45892  32452 S   0.0   0.6   0:00.00 containerd --config /var/run/docker/containerd/conta
  463 root     20   0  1751M 45892  32452 S   0.0   0.6   0:01.35 containerd --config /var/run/docker/containerd/conta
  469 root     20   0  1751M 45892  32452 S   0.0   0.6   0:01.62 containerd --config /var/run/docker/containerd/conta
  478 root     20   0  1751M 45892  32452 S   0.0   0.6   0:00.00 containerd --config /var/run/docker/containerd/conta
  480 root     20   0  1974M 84472  61216 S   0.0   1.0   0:00.20 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 -
  481 root     20   0  1974M 84472  61216 S   0.0   1.0   0:00.00 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 -
  482 root     20   0  1974M 84472  61216 S   0.0   1.0   0:00.20 /usr/bin/dockerd -p /var/run/docker.pid --mtu=1460 -
  650 root     20   0    2696    1648   1552 S   0.0   0.0   0:00.00 sleep infinity
  652 root     10 -10   13996    9600   8220 S   0.0   0.1   0:00.02 sshd: akarshana2428 [priv]
  660 root     10 -10   13996    9688   8316 S   0.0   0.1   0:00.02 sshd: akarshana2428 [priv]
  665 akarshana2 10 -10   14256    6252   4592 S   0.0   0.1   0:00.00 sshd: akarshana2428@pts/0
  666 akarshana2 10 -10    7340    3816   3568 S   0.0   0.0   0:00.00 bash -c (sudo touch /var/run/google/devshell/37899 &
F1[F2[F3[F4[F5[F6[F7[F8[F9[F10[F11
```

IOSTAT:

```
akarshana2428@cloudshell:~$ iostat
Linux 6.6.137+ (cs-897017188372-default)      05/14/2026      _x86_64_      (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
            1.25    0.00    0.66    0.06    0.01   98.03

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
dm-0              0.70        38.08        0.00        0.00    438640        0        0
loop0            0.01         0.06        0.00        0.00         673         8        0
sda              4.89       146.64       36.08        0.00   1689290   415621        0
sdb              0.13       11.50        4.82        0.00    132538    55488        0
```

VMSTAT:

```
akarshana2428@cloudshell:~$ vmstat
procs -----memory----- ---swap-- ----io---- -system-- -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi   bo    in   cs   us   sy   id   wa   st   gu
 2   0       0 4353424 90892 1538272    0    0   157   41  487    5   1   1 98   0   0   0
```

SAR:

```
akarshana2428@cloudshell:~$ sar -u 1 5
Linux 6.6.137+ (cs-897017188372-default)      05/14/2026      _x86_64_      (2 CPU)

05:44:38 PM   CPU      %user   %nice   %system   %iowait   %steal   %idle
05:44:39 PM   all       1.50    0.00    0.00    0.00    0.50   98.00
05:44:40 PM   all       0.50    0.00    1.49    0.00    0.00   98.02
05:44:41 PM   all       0.51    0.00    0.00    0.00    0.00   99.49
05:44:42 PM   all       1.00    0.00    0.50    0.00    0.00   98.51
05:44:43 PM   all       0.50    0.00    1.50    0.00    0.00   98.00
Average:      all       0.80    0.00    0.70    0.00    0.10   98.40
```

FREE:

```
akarshana2428@cloudshell:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.8Gi        2.3Gi        4.1Gi        1.1Mi        1.6Gi        5.5Gi
Swap:          0B           0B           0B
```

MPSTAT:

```
akarshana2428@cloudshell:~$ mpstat
Linux 6.6.137+ (cs-897017188372-default)      05/14/2026      _x86_64_      (2 CPU)

05:45:33 PM   CPU      %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %idle
05:45:33 PM   all       1.26    0.00    0.62    0.06    0.00    0.05    0.01    0.00    0.00   98.01
```

UPTIME:

```
akarshana2428@cloudshell:~$ uptime
17:46:02 up 3:17, 0 user, load average: 0.01, 0.07, 0.08
```

DSTAT:

```
akarshana2428@cloudshell:~$ dstat -cdnm --nouupdate
Error Parsing ASCII PMNS: Cannot open "/var/lib/pcp/pmns/root"
Failed to register derived metric:
PMNS not accessible <pcp.pmapi.c_uint_Array_1 object at 0x7ff32288dd0>.
```

GLANCES:

```
CPU [ 5.9%] CPU 1 5.9% idle 98.5% ctx_sw 475 MEM - 30.8% SWAP - 0.0% LOAD - 2core
user 1.0% irq 0.0% inter 344 total 7.76G total 0 1 min 0.26
system 0.5% nice 0.0% sw_int 194 used 2.39G used 0 5 min 0.21
swap [ 0.0%] iowait 0.0% steal 0.0% free 5.37G free 0 15 min 0.13

NETWORK Rx/s Tx/s TASKS 56 (148 thr), 1 run, 55 slp, 0 oth Threads sorted automatically by CPU consumption
eth0 15Kb 48Kb CPU% MEM% VIRT RES PID USER TIME+ THR NI S R/s W/s
lo 38Kb 38Kb >1.5 0.6 202M 46.1M 7247 akarshana 0:01 1 -10 R 0 0 gvyshon3 /usr/bin/g
DefaultGateway 11ms 0.5 0.1 8.93M 4.57M 687 akarshana 0:00 1 -10 S 0 0 tmux new-session -
0.0 11.7 42.6G 952M 1451 akarshana 0:53 13 0 S ? ? node --dns-result-
0.0 7.6 1.77G 602M 1513 akarshana 0:09 11 0 S ? ? node /google/devsh
sda 0 387K 0.0 1.3 11.3G 104M 911 akarshana 0:06 11 0 S ? ? node /google/devsh
sda1 0 387K 0.0 1.0 1.93G 82.5M 320 root 0:01 10 0 S ? ? dockerd -p /var/ru
sda2 0 0 0.0 0.9 1.79G 70.5M 2531 akarshana 0:00 8 0 S 0 0 cloudcode cli duet
sda3 0 0 0.0 0.8 993M 60.9M 1498 akarshana 0:00 7 0 S ? ? node /google/devsh
sda4 0 0 0.0 0.6 1024M 60.9M 1463 akarshana 0:00 11 0 S ? ? gude /google/devsh
```

9. USER MANAGEMENT

USERADD:

```
akarshana2428@cloudshell:~$ sudo adduser demo
info: Adding user `demo' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `demo' (1001) ...
info: Adding new user `demo' (1001) with group `demo (1001)' ...
warn: The home directory `/home/demo' already exists. Not touching this d
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for demo
Enter the new value, or press ENTER for the default
    Full Name []: AKAR
    Room Number []: 21
    Work Phone []: 1234567890
    Home Phone []: 8765432111
    Other []: 0
Is the information correct? [Y/n] Y
info: Adding new user `demo' to supplemental / extra groups `users' ...
info: Adding user `demo' to group `users' ...
```

USERDEL:

```
akarshana2428@cloudshell:~$ sudo deluser demo
info: Removing crontab ...
info: Removing user `demo' ...
```

USERMOD:

```
akarshana2428@cloudshell:~$ sudo usermod -aG sudo demo
```

PASSWD:

```
akarshana2428@cloudshell:~$ sudo passwd demo
New password:
Retype new password:
passwd: password updated successfully
```

CHAGE:

```
akarshana2428@cloudshell:~$ sudo chage -l demo
Last password change                : May 14, 2026
Password expires                    : never
Password inactive                    : never
Account expires                     : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
```

GROUPS:

```
akarshana2428@cloudshell:~$ groups demo
demo : demo sudo
```

GROUPADD:

```
akarshana2428@cloudshell:~$ sudo groupadd dev
akarshana2428@cloudshell:~$ getent group dev
dev:x:1003:
```

GROUPDEL:

```
akarshana2428@cloudshell:~$ sudo groupdel dev
```

NEWGRP:

```
akarshana2428@cloudshell:~$ sudo groupadd developers
```

```
akarshana2428@cloudshell:~$ groups
dev adm sudo docker akarshana2428 developers
```

SU:

```
akarshana2428@cloudshell:~$ sudo adduser john
info: Adding user `john' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `john' (1005) ...
info: Adding new user `john' (1005) with group `john (1005)' ...
info: Creating home directory `/home/john' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for john
Enter the new value, or press ENTER for the default
  Full Name []: AKARSH
  Room Number []: 22
  Work Phone []: 12345667
  Home Phone []: 76543211
  Other []: N
Is the information correct? [Y/n] Y
info: Adding new user `john' to supplemental / extra groups `users' ...
info: Adding user `john' to group `users' ...
```

10. COMPRESSION AND ARCHIVING

```
akarshana2428@cloudshell:~$ touch file.txt
```

BZIP2

```
akarshana2428@cloudshell:~$ bzip2 file.txt
```

BUNZIP2:

```
akarshana2428@cloudshell:~$ bunzip2 file.txt.bz2
```

XZ:

```
akarshana2428@cloudshell:~$ touch file.txt
akarshana2428@cloudshell:~$ xz file.txt
```

UNXZ:

```
akarshana2428@cloudshell:~$ unxz file.txt.xz
```

TAR:

```
akarshana2428@cloudshell:~$ mkdir folder
akarshana2428@cloudshell:~$ tar -cvf archive.tar folder/
folder/
```

ZIP:

```
akarshana2428@cloudshell:~$ touch file.txt
akarshana2428@cloudshell:~$ zip archive.zip file.txt
  adding: file.txt (stored 0%)
```

UNZIP:

```
akarshana2428@cloudshell:~$ unzip archive.zip
Archive:  archive.zip
replace file.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: Y
extracting: file.txt
```

LZMA:

```
akarshana2428@cloudshell:~$ touch file.txt
akarshana2428@cloudshell:~$ lzma file.txt
```

LZOP:

```
akarshana2428@cloudshell:~$ touch file.txt
akarshana2428@cloudshell:~$ lzop file.txt
```

UNLZOP:

```
akarshana2428@cloudshell:~$ unlzop file.txt.lzo
```

